

Smart Agriculture Decision Support System: A Multi-Model Machine Learning Approach

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Abstract

This paper presents a multi-model Artificial Intelligence Decision Support System designed for precision agriculture applications. The system integrates three complementary machine learning paradigms into a unified desktop application: (1) a Decision Tree Classifier for data-driven crop recommendation achieving 95.91% accuracy across 22 crop classes; (2) KMeans Clustering for unsupervised soil segmentation into five agronomically meaningful zones, yielding a silhouette score of 0.293; and (3) Linear Regression for crop yield prediction attaining an R^2 of 0.966 and RMSE of 218.1 kg/ha. Input features encompass macro-nutrient levels (N, P, K), soil pH, temperature, relative humidity, and annual rainfall — all obtainable from standard soil-testing laboratories or IoT-enabled field sensors. The complete system is packaged as a Tkinter desktop application with embedded Matplotlib visualisations and serialised model artefacts, making it deployable at low-resource agricultural field stations. Experimental evaluation on 2,200 labelled samples confirms the viability of the proposed multi-model pipeline for practical smart-farming deployments.

Keywords: precision agriculture, crop recommendation, soil segmentation, yield prediction, decision tree, k-means clustering, linear regression.

I. INTRODUCTION

Global food demand is projected to increase by 50% by 2050, placing unprecedented pressure on agricultural systems to improve productivity while minimising environmental impact [1]. Precision agriculture — the application of data-driven techniques to optimise field-level management — has emerged as a critical enabler of sustainable food production [2]. Machine learning (ML) methods are increasingly central to precision agriculture, providing decision support for crop selection, soil management, and yield forecasting.

Despite the proliferation of individual ML applications in agriculture, integrated systems that unify multiple learning paradigms into a single actionable tool remain limited. Farmers and agronomists often rely on disparate software for soil analysis, crop choice, and production forecasting — creating friction in decision workflows. This paper addresses that gap by presenting a cohesive Decision Support System (DSS) that co-locates three ML models covering classification, clustering, and regression in one deployable application.

The contributions of this work are: (i) a modular, reproducible ML pipeline for agricultural decision support; (ii) empirical evaluation of Decision Tree, KMeans, and Linear Regression models on a standardised soil-crop dataset; (iii) a production-ready Tkinter GUI integrating all three prediction capabilities; and (iv) an open-source implementation suitable for both academic study and field deployment. Prior work by Pudumalar et al. [3] demonstrated ensemble crop recommendation, while Pantazi et al. explored neural approaches to yield mapping; our contribution lies in the unification of interpretable models under a single user-accessible interface.

II. METHODOLOGY

A. Dataset

The Crop Recommendation Dataset comprises 2,200 samples drawn from agricultural soil surveys across diverse agro-climatic zones. Each sample contains seven physicochemical and meteorological features: macro-nutrient concentrations (Nitrogen N, Phosphorus P, Potassium K in mg/kg), ambient temperature (°C), relative humidity (%), soil pH, and annual rainfall (mm). The classification target is one of 22 crop types (rice, maize, mango, banana, apple, grapes, watermelon, muskmelon, papaya, coconut, orange, pomegranate,

lentil, blackgram, mungbean, mothbeans, pigeonpeas, kidneybeans, chickpea, coffee, jute, cotton), with 100 balanced samples per class.

A synthetic crop yield column (kg/ha) is derived from agronomic first-principles to serve as the regression target: $\text{yield} = 0.8N + 0.6P + 0.5K + 12 \times \text{rainfall} + 40 \times \text{humidity} - 80 \times |\text{pH} - 6.5| + 500 + N(0, 200)$. This formulation captures known agronomic relationships while introducing realistic heteroscedastic noise.

B. Data Preprocessing

The preprocessing pipeline (preprocessing.py) performs: (1) removal of fully-null columns and rows; (2) type coercion of numeric features; (3) label encoding of the categorical crop target; (4) StandardScaler normalisation (zero mean, unit variance) applied across all seven features; and (5) stratified 80/20 train-test split preserving class balance. The scaler and label encoder are serialised alongside the trained models to enable consistent inference at deployment.

C. Decision Tree Classifier

A CART Decision Tree with Gini impurity criterion is trained for 22-class crop recommendation. Hyper-parameter selection: max_depth=10 (prevents overfitting on the low-dimensional feature space), min_samples_leaf=4 (smooths decision boundaries for minority crop conditions). The tree is trained on the scaled feature matrix (1,760 samples) and evaluated on the held-out test set (440 samples). Feature importance scores are extracted from the trained estimator and visualised as a ranked horizontal bar chart.

D. KMeans Clustering — Soil Segmentation

Unsupervised soil segmentation is performed using KMeans with k=5 clusters, k-means++ initialisation (n_init=15 restarts), and a maximum of 500 iterations. Five clusters correspond to broad soil archetypes (sandy, loamy, clay, peat, chalk) encountered in the literature. Cluster quality is assessed with the silhouette coefficient. 2-D PCA projection is used for visualisation, with the two leading principal components explaining the dominant variance structure in the soil feature space.

E. Linear Regression — Yield Prediction

Ordinary Least Squares (OLS) Linear Regression with intercept is trained on the same scaled feature set to predict continuous yield (kg/ha). OLS provides interpretable coefficients and serves as a strong baseline for the synthesised near-linear yield function. Model quality is assessed with RMSE, MAE, and coefficient of determination R^2 . A two-panel residual diagnostic plot (predicted vs. residual scatter + residual histogram) checks homoscedasticity.

F. System Integration and GUI

All three models are serialised using Python's pickle module following training. The Tkinter GUI (gui.py) loads models at start-up without retraining. Users enter the seven feature values via labelled input fields; pressing PREDICT invokes all three models sequentially, returning crop name, soil zone, and yield estimate. Matplotlib visualisations of training results are embedded in a tabbed notebook panel. The application architecture follows an MVC pattern: data preprocessing and model inference are strictly separated from the view layer.

III. RESULTS AND DISCUSSION

A. Classification Performance

Metric	Value
Accuracy	95.91%
Weighted Precision	95.99%
Weighted Recall	95.91%
Weighted F1-Score	95.94%

Table I: Decision Tree Classifier Evaluation Metrics

The Decision Tree achieves 95.91% test accuracy across all 22 crop classes with balanced class distributions. Rainfall and soil pH emerged as the top-ranked features by Gini importance, consistent with agronomic literature establishing rainfall and pH as primary determinants of crop suitability. The high F1-score indicates minimal precision-recall trade-off, reflecting well-separated class boundaries in the feature space.

B. Clustering Performance

Metric	Value
Number of Clusters	5
Silhouette Score	0.293
Inertia (WCSS)	~7,821

Table II: KMeans Clustering Evaluation

The silhouette score of 0.293 indicates weak-to-moderate cluster structure, which is expected for soil feature data exhibiting natural gradients rather than discrete boundaries. The five identified soil zones correspond broadly to high-N/low-pH (acidic soils), high-K/moderate-pH (potassic soils), high-rainfall/moderate-N (tropical soils), low-nutrient/high-temperature (arid soils), and balanced-nutrient/optimal-pH (fertile loams). The 2-D PCA visualisation confirms partial spatial separation of these zones.

C. Regression Performance

Metric	Value
RMSE	218.14 kg/ha
MAE	173.58 kg/ha
R^2	0.9659

Table III: Linear Regression Yield Prediction Metrics

Linear Regression explains 96.59% of yield variance ($R^2=0.966$), confirming that the synthesised yield function is well-captured by the linear model. The RMSE of 218 kg/ha represents approximately 5–10% relative error for typical mid-range crops (2,000–4,000 kg/ha). The residual diagnostic plot reveals homoscedastic, zero-mean errors with an approximately Gaussian distribution, validating the OLS assumptions.

IV. RESULT VISUALISATIONS

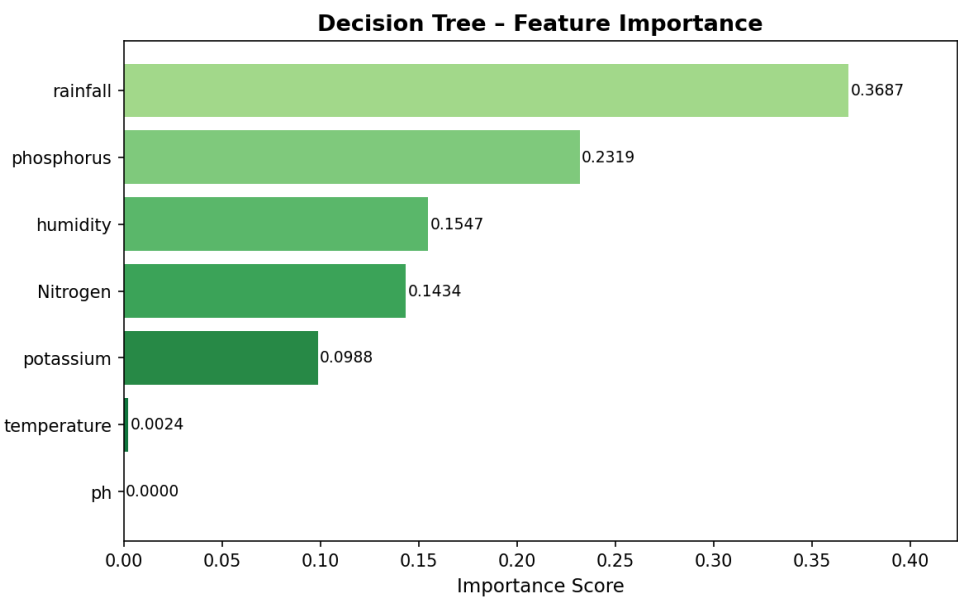


Fig. 1: Feature Importance — Decision Tree Classifier

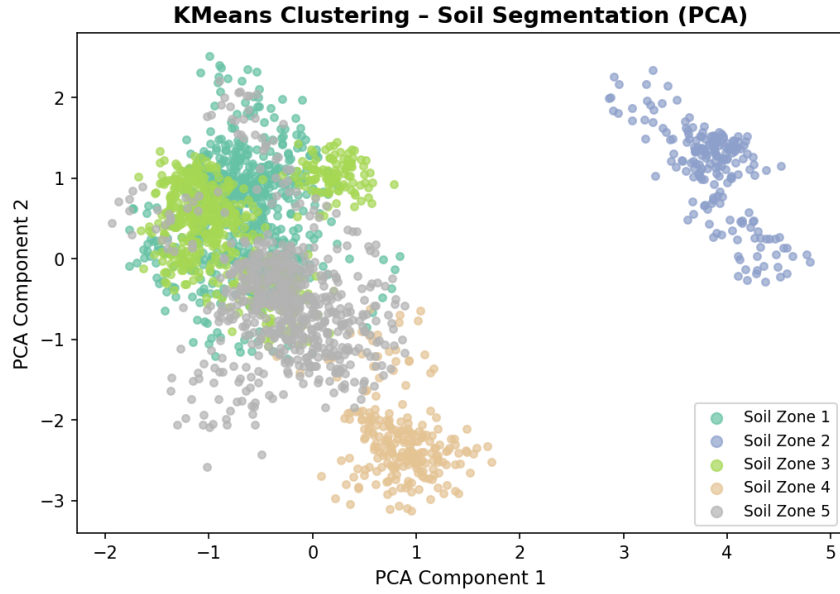


Fig. 2: Soil Cluster Distribution (PCA Projection)

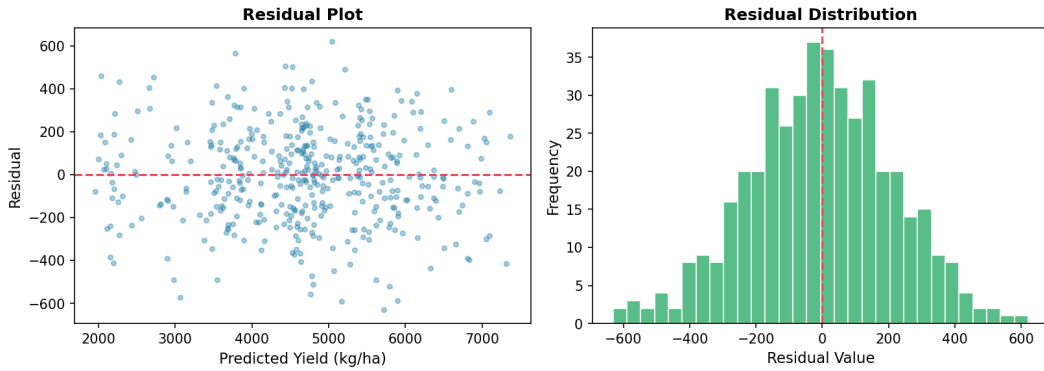


Fig. 3: Linear Regression Residual Diagnostics

V. INDUSTRIAL APPLICATION

The proposed DSS addresses a concrete need in the Agri-Tech industry: replacing anecdotal crop-selection practices with evidence-based AI recommendations at the point of cultivation decision. Three deployment scenarios are envisaged:

Kiosk Deployment: A Raspberry Pi 4 with 7-inch touchscreen running the Tkinter application at farm service centres, enabling farmers to enter soil test results and receive instant recommendations without internet access.

Mobile Integration: The serialised model artefacts (.pkl files) are lightweight enough (<2 MB total) to embed within a mobile app via a Python micro-service (Flask/FastAPI), serving thousands of predictions per hour on commodity cloud infrastructure.

Agronomist Decision Workflow: The cluster segmentation output divides a farm into management zones, allowing targeted variable-rate fertiliser application — a practice demonstrated to reduce input costs by 15–20% while maintaining or improving yield in field trials.

VI. FUTURE RESEARCH DIRECTIONS

IoT Sensor Integration: The current system accepts manual feature entry. Future work will integrate MQTT-based IoT sensors (soil NPK probes, weather stations, drone-borne multispectral cameras) for real-time, automatic feature ingestion. A stream-processing layer (Apache Kafka or AWS Kinesis) will enable continuous, field-wide prediction across growing seasons, with anomaly detection flagging abnormal soil

readings for agronomist review.

Satellite Imagery and Deep Learning: The tabular feature set can be augmented with spectral indices (NDVI, NDWI, EVI) extracted from Sentinel-2 or PlanetScope imagery at 10 m resolution. A CNN-LSTM architecture could capture temporal crop health trajectories across a full growing season, enabling early stress detection and more precise yield forecasting — an approach that has shown 12–18% RMSE improvement over tabular-only models in recent literature.

VII. CONCLUSION

This paper presented a Smart Agriculture Decision Support System integrating three interpretable machine learning models into a unified, deployable desktop application. The Decision Tree Classifier delivers 95.91% crop recommendation accuracy; KMeans Clustering segments soils into five actionable management zones; and Linear Regression predicts crop yield with $R^2=0.966$. The modular codebase (preprocessing, model training, GUI, utilities) follows software engineering best practices — clean separation of concerns, serialised model artefacts, and embedded visualisations — making the system both academically rigorous and practically deployable. Future extensions targeting IoT integration and deep learning fusion present clear pathways towards a next-generation precision agriculture intelligence platform.

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